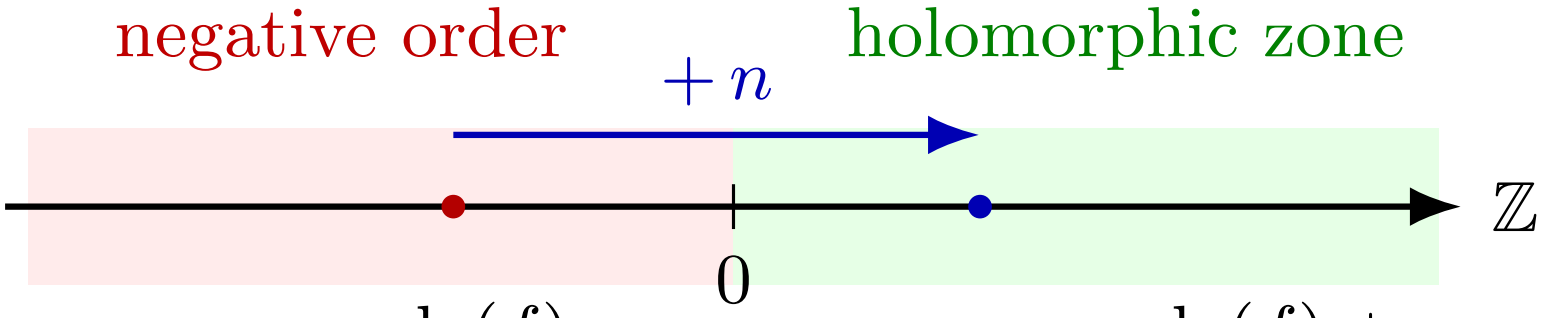


$$z^n f \text{ holomorphic at } p \iff \mathrm{ord}_p(z^n f) \geq 0$$

$$\mathrm{ord}_p(z^n f) = \mathrm{ord}_p(f) + n \geq 0.$$